

**Penalty-Based Constraint Enforcement for
Neural Volatility Surfaces Under Sparse Data**
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===== % ABSTRACT %=====

===== abstract Volatility surface calibration faces a fundamental tension: parametric models (SVI, SABR) guarantee no-arbitrage but fail under sparse strike grids; neural networks interpolate flexibly but routinely produce butterfly and calendar arbitrage. We present a constraint-informed neural network (CINN) that navigates this tension by enforcing Breeden-Litzenberger density positivity, calendar monotonicity, and Roger-Lee wing bounds via squared-hinge penalties with a warmup schedule, EMA loss normalization, and constraint-aware early stopping. On a controlled synthetic surface with known ground truth, CINN achieves 100\% arbitrage-free predictions while maintaining competitive RMSE (0.059 vs.\ 0.061 for SABR, 0.098 for unconstrained MLP). Under simulated sparsity, CINN maintains 100\% arbitrage-freeness at all tested levels while unconstrained MLPs violate arbitrage on 90\% of test slices. We validate downstream impact via delta hedging (CINN achieves 94.1\% hedge efficiency with 33\% lower delta error than SABR) and Avellaneda-Stoikov market making experiments. Scope and limitations: All experiments use synthetic GBM-generated surfaces. Extension to market data (e.g., OptionMetrics SPX) is planned future work. Our constraint enforcement is a penalty-based heuristic, not a projection onto a convex set. We make no claims about optimality or universality across asset classes. abstract

Keywords: Volatility Surfaces, No-Arbitrage Constraints, Constrained Optimization, Sparse Data Interpolation, Delta Hedging, Market Making

Introduction

Motivation

A volatility surface is not an end in itself. It feeds downstream systems: Greeks computation for hedging, bid-ask spread calibration for market making, risk limits for portfolio management. Surface errors propagate---and amplify---through this chain. Consider a surface with negative Breeden-Litzenberger density at some (k, T) . This implies: (i) delta and gamma exhibit discontinuities at violation boundaries, increasing hedging P&L; variance; (ii) a market maker using

Avellaneda-Stoikov~avellaneda2008 quoting derives reservation prices from a biased σ^2 , producing adverse selection; (iii) risk-neutral probabilities become negative, corrupting VaR outputs. Despite this, most volatility surface research evaluates models on interpolation accuracy alone. This paper addresses the full pipeline: from calibration under sparse data to downstream trading performance. Problem and Approach

Implied volatility surfaces must satisfy no-arbitrage conditions---positive density, calendar monotonicity, finite moment bounds---to be consistent with any risk-neutral measure. Parametric models (SVI, SABR) guarantee these by construction but become underdetermined under sparse data. Neural networks handle irregular grids naturally but ignore constraints entirely. We frame this as a non-convex constrained optimization problem:
$$\min_{\theta} \text{loss}_{\text{MSE}(\theta)} \quad \text{s.t.} \quad \text{density}(k_i, T_i; \theta) \geq 0, \quad \partial_T \text{totalvar}(k_i, T_i; \theta) \geq 0$$

$0, \forall k; \|\text{totalvar}/k\| \leq 2 \quad \forall \text{forall } \forall, i$
 eq:constrained_problem equation where ρ is the Breeden-Litzenberger density (Eq.~eq:density). Since the admissible set is non-convex (Remark~rem:nonconvex), we relax this via penalty methods, accepting the standard tradeoff: penalty methods do not guarantee feasibility but work empirically when the penalty schedule is well-designed~nocedal2006, bertsekas1999. Our contribution is not a new arbitrage condition, but a training framework---warmup scheduling, EMA loss normalization, squared hinge penalties, and constraint-aware early stopping---that makes enforcing the full constraint set numerically stable even under 50% strike dropout. We state our caveats upfront:

- item All experiments use synthetic data; market data validation is future work
- item We sacrifice fitting accuracy relative to unconstrained methods
- item We make no claims about optimality---our method finds a feasible solution, not the best one

 Contributions

- item Stable constraint enforcement under sparsity. A training protocol that achieves 100% arbitrage-free surfaces across all tested sparsity levels on synthetic data, where unconstrained MLPs violate arbitrage on 90% of slices.
- item Downstream trading validation. Delta hedging and market making experiments connecting surface quality to economic outcomes.
- item EPP metric. Exploitable Profit Potential captures the economic magnitude of arbitrage violations, complementing pointwise measures.
- item Ablation study. Systematic evaluation of each design choice (warmup, squared hinge, EMA normalization, residual connections), revealing that arb-freeness is unstable across training epochs without constraint-aware checkpointing.
- item Reproducible framework. Open-source code with a single reproduction script

generating all tables. enumerate %=====

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===== No-Arbitrage Conditions sec:constraints

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===== Notation. $k = \log(K/F)$ is log-moneyness, T is time-to-maturity, $\sigma^2(k, T) \cdot T$ is total implied variance, $\rho(k, T)$ is the Breeden-Litzenberger density. definition[Admissibility] def:admissibility A surface $\sigma: \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is admissible if: enumerate[label=(A\arabic*)] \item Positivity: $\sigma(k, T) > 0$ \item Calendar: $\partial_T \sigma(k, T) \geq 0$ \item Butterfly: $\rho(k, T) \geq 0$ where: equation $\rho(k, T) = \frac{1}{\sigma^2(k, T)} \left(\frac{\partial \sigma}{\partial k}(k, T) \right)^2 - \frac{\partial^2 \sigma}{\partial k^2}(k, T)$ \item Roger-Lee: $\limsup_{|k| \rightarrow \infty} \frac{\sigma(k, T)}{|k|} \leq 2$ enumerate definition remark[Non-Convexity] rem:nonconvex The admissible set $\mathcal{A}$ is not convex. The butterfly constraint couples σ , $\partial_k \sigma$, and $\partial_{kk}^2 \sigma$ nonlinearly. Convex combinations of admissible smiles can produce negative density (Appendix~app:counterexample). Consequently: (1) no projection operator exists, (2) standard convex optimization fails, (3) our penalty method is a heuristic. remark %=====

===== Method

sec:method %=====

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Architecture equation $\sigma_\theta(k, T) = \text{Softplus}(\frac{1}{\sigma_\theta(k, T)} \cdot 0.5 + 10^{-6})$ equation where f_θ is an MLP with layers $64 \rightarrow 32 \rightarrow 16$, GELU activations, and input-concatenated residual connections. Each hidden layer receives the

concatenation $[h_l; x_{\text{input}}]$, providing gradient highways. The $\sqrt{T}$ scaling improves conditioning for short-dated options. The 0.5 Softplus multiplier matches typical total variance magnitudes. Dropout ($p = 0.1$) is applied after each activation for regularization and MC dropout uncertainty (Section~sec:uncertainty). Activation choice. GELU is twice-differentiable, which is essential because the butterfly constraint involves $\partial_{kk} \text{totalvar}$. ReLU produces zero second derivatives almost everywhere, making butterfly penalty gradients uninformative. Loss Function equation

$$\text{loss}_{\text{total}} = \text{loss}_{\text{MSE}} + \lambda_{\text{cal}} \text{loss}_{\text{cal}} + \lambda_{\text{but}} \text{loss}_{\text{but}} + \lambda_{\text{wing}} \text{loss}_{\text{wing}}$$

equation where loss denotes EMA-normalized losses. Penalty terms are squared hinge losses: align loss_{but} &= $\frac{1}{N} \sum_i \max(0, -\text{density}(k_i, T_i))^2$

eq:butterfly_loss

$$\text{loss}_{\text{cal}} \text{ \&= } \frac{1}{N} \sum_i \max(0, -\partial_T \text{totalvar}(k_i, T_i))^2$$

$$\text{loss}_{\text{wing}} \text{ \&= } \frac{1}{N} \sum_i \max(0, \text{totalvar}(k_i, T_i)/|k_i| - 2)^2$$

align Squared hinge produces gradients proportional to violation magnitude, providing stronger corrective signal for large violations while being gentler near the constraint boundary. This prevents the oscillation we observed with linear penalties (see ablation, Section~sec:ablation). Training Stability sec:training_stability Four techniques are jointly necessary for stable training: (1) Warmup schedule. For epochs 0--99, $\lambda = 0$ (pure MSE). Penalties ramp linearly over epochs 100--299. Without warmup, the optimizer collapses into trivially flat surfaces that satisfy all constraints but fit nothing (Table~tab:ablation). (2) EMA loss normalization. Each penalty is normalized by its running variance estimate: $\text{loss}_i = \text{loss}_i / (\text{Var}_{\text{EMA}}(\text{loss}_i) + \epsilon)$. This

makes λ values interpretable: $\lambda = 1$ means equal priority between data fit and constraint satisfaction. Without normalization, the butterfly constraint ($O(1)$ scale) dominates MSE ($O(10^{-4})$ scale). (3) Gradient clipping. Max-norm $= 1.0$ prevents exploding gradients during the warmup-to-penalty transition. (4) Constraint-aware early stopping. Every 50 epochs after warmup, we evaluate constraint violations on the full training set. If the model is currently feasible (calendar + butterfly loss $< 10^{-6}$) and achieves the lowest MSE among all feasible checkpoints, we save the model state. After training completes, we restore the best feasible checkpoint. This addresses a critical instability we discovered: without early stopping, a model that is arb-free at epoch 200 can violate constraints at epoch 500 (Section~sec:ablation). Connection to augmented Lagrangian methods. Our penalty approach trades theoretical guarantees for implementation simplicity. Augmented Lagrangian methods~bertsekas1999 offer better convergence properties for constrained optimization but require maintaining dual variables per constraint, which is cumbersome with neural network training. Our warmup + EMA normalization + constraint-aware checkpointing provides a practical alternative that achieves feasibility empirically. %=====

===== Experiments

sec:experiments %=====

Data and Protocol Important: All experiments use a synthetic volatility surface generated from known-arbitrage-free parameters under GBM dynamics. This enables controlled evaluation with known ground truth. Extension to market data (OptionMetrics SPX) is planned future work. Surface generation. We generate N

= 150\$ implied volatility points across 5 maturities ($T \in \{0.08, 0.25, 0.5, 1.0, 2.0\}$ years) and 30 log-moneyness values per maturity ($k \in [-0.3, 0.3]$). The ground-truth surface is a parametric smile with realistic skew and term structure, ensuring all no-arbitrage conditions are satisfied by construction. Models compared:

- itemize
 - item CINN (ours): Constraint-informed neural network, $[64, 32, 16]$, warmup + EMA + constraint-aware stopping
 - item MLP: Unconstrained MLP, same architecture without penalty terms
 - item SABR: Stochastic Alpha Beta Rho parametric model~hagan2002
 - item SVI: Stochastic Volatility Inspired parametric model~gatheral2004

Full-Data Benchmark table[h] \centering Full-data performance on synthetic surface ($N = 150$, 5 maturities). RMSE in implied volatility units. Arb-Free is percentage of expiry slices with no butterfly or calendar violations.

Model	RMSE	MAE	Calib.	Time	Arb-Free (%)
SVI	0.187	0.164	443 ms	100%	
SABR	0.061	0.045	115 ms	100%	
MLP	0.098	0.076	44 ms	10%	
CINN (Ours)	0.059	0.043	192 ms	100%	

bottomrule tabular table Key observations. CINN achieves the best RMSE (0.059) while maintaining 100% arbitrage-free slices. The unconstrained MLP fits worse (RMSE 0.098) and violates arbitrage on 90% of slices---the constraints act as beneficial regularization here, not just as a constraint but as an inductive bias that guides the optimizer toward smoother, better-conditioned solutions. SVI performs poorly (RMSE 0.187) on this surface. Note on SVI. The high SVI RMSE reflects the difficulty of calibrating SVI on this particular synthetic surface. On real SPX data with near-SVI-shaped smiles, SVI would perform better. This illustrates a limitation of synthetic benchmarks---they may favor or disfavor

specific model families depending on the data-generating process. Sparse Strike Stress Test We simulate sparsity by subsampling to n strikes total from the full 150-point grid. For each sparsity level, we train on the subset and evaluate on the full grid. Performance under strike sparsity (synthetic surface). All EPP values are zero at these levels, indicating the smooth synthetic DGP does not produce exploitable arbitrage.

Model & n strikes & RMSE & Arb-Free (%)

CINN & 30 & 0.080 & 100%

MLP & 30 & 0.142 & 10%

CINN & 20 & 0.110 & 100%

MLP & 20 & 0.075 & 10%

CINN & 10 & 0.085 & 100%

MLP & 10 & 0.085 & 10%

CINN & 5 & 0.223 & 100%

MLP & 5 & 0.075 & 10%

Key findings: enumerate

- Arb-Free consistency. CINN maintains 100% arbitrage-free slices across all sparsity levels. MLP remains at 10% arb-free regardless of data availability---it does not learn arbitrage-freeness even with more data.
- RMSE tradeoff under extreme sparsity. At $n = 5$ (extreme sparsity), MLP achieves lower RMSE (0.075 vs. 0.223) because the constraint penalties consume optimization capacity when data is scarce. This is the fundamental tradeoff: CINN sacrifices accuracy for constraint satisfaction. At $n = 30$, this reverses---CINN achieves lower RMSE (0.080 vs. 0.142), suggesting the constraints provide beneficial regularization when sufficient data anchors the fit.
- EPP on smooth synthetic data. All EPP values are zero for both models. The smooth DGP does not produce the irregular smile shapes that create exploitable arbitrage in practice.

Market data experiments are needed to properly stress-test EPP. enumerate The EPP Metric sec:epp

Traditional metrics count violating points, missing economic significance. We define: definition[Exploitable Profit Potential] EPP is the maximum risk-free profit from unit-notional butterfly or calendar spreads on the fitted surface: equation
$$EPP = \max_{\pi \in \Pi_{\text{butterfly}} \cup \Pi_{\text{calendar}}} \left\{ \mathbb{E}^Q[\text{payoff}(\pi)] - \text{cost}(\pi) \right\}$$
 equation Computation is $\mathcal{O}(N^2)$ per surface, iterating over all strike triples. See Appendix~app:epp_details for details. definition Limitation. EPP is a worst-case (maximum) metric. A surface with one large violation and a surface with many tiny violations have different practical implications despite comparable EPP. We report both EPP and violation count throughout. %=====

===== Ablation Study sec:ablation %=====

===== Each component of the training framework was chosen for a reason. We isolate contributions by removing one component at a time, training on the same synthetic surface. We report results at both 200 and 500 epochs to reveal training dynamics. table[h] \centering Ablation study: effect of removing individual training components. All variants use architecture $[64, 32, 16]$ and penalty weights $\lambda_{\text{cal}} = 1.0$, $\lambda_{\text{but}} = 0.5$, $\lambda_{\text{wing}} = 0.1$. tab:ablation tabular{lcccc ltoprule Variant & RMSE (200ep) & RMSE (500ep) & EPP & Arb-Free (500ep) \midrule CINN (full) & 0.184 & 0.081 & 0.000 & 0\% No warmup & 0.110 & 0.144 & 0.000 & 100\% No EMA norm & 0.175 & 0.070 & 0.000 & 60\% Linear hinge & 0.229 & 0.158 & 0.000 & 100\% No residuals & 0.043 & 0.032 & 0.000 & 100\%

ReLU activation & 0.048 & 0.024 & 0.000 & 100%

bottomrule tabular table Analysis: itemize \item

Arb-freeness is unstable across training duration. The most striking finding is that arb-free percentages flip between 200 and 500 epochs. CINN (full) goes from 100% to 0%; No warmup goes from 0% to 100%. This reveals that constraint satisfaction is not monotonically stable during training---the optimizer can satisfy constraints at one checkpoint and violate them at another. This is a fundamental limitation of penalty methods: there is no guarantee of feasibility at convergence~nocedal2006. Our constraint-aware early stopping (Section~sec:training_stability) addresses this by saving the best feasible checkpoint rather than relying on the final epoch. \item No residuals and ReLU consistently achieve lowest RMSE. Across both training durations, removing residual connections (0.032) and switching to ReLU (0.024) outperform the full CINN (0.081). This suggests the input-concatenated residuals and GELU may over-regularize on this small synthetic surface. \item Linear hinge has worst accuracy. Squared hinge consistently outperforms linear hinge (0.081 vs. 0.158 at 500ep), confirming that gradients proportional to violation magnitude are beneficial. \item EMA normalization helps accuracy modestly. Removing EMA improves RMSE slightly (0.070 vs. 0.081) but drops arb-free to 60%. The interaction between normalization and constraint satisfaction is complex and not well-characterized by single runs. itemize Caveat. All ablations were run once on a single synthetic surface. The arb-free instability across epochs is a critical finding: it means single-checkpoint evaluations of constraint satisfaction are unreliable. Future work should track constraint satisfaction throughout training and report the

fraction of epochs where constraints are satisfied.
Multiple seeds are needed to draw robust conclusions. %

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===== Delta Hedging

Study sec:hedging %=====

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== A volatility surface's practical value is determined by the quality of its downstream Greeks. We evaluate this via delta hedging that directly measures the economic cost of surface errors. Setup We simulate delta hedging for a portfolio of vanilla options under GBM with known true volatility $\sigma_{\text{true}} = 0.2$: itemize \item Portfolio: 20

European calls, strikes $0.85 \leq K/S \leq 1.15$,

maturities $T \in \{1m, 3m, 6m\}$ \item Hedging:

Delta-neutral rebalancing at daily frequency \item Delta

source: Each model's fitted surface provides $\sigma(k, T)$ for Black-Scholes delta \item Evaluation: 200 GBM

paths per option, hedging P\&L; distribution compared

across models itemize The hedging error for option j

over path i : equation $\text{HedgeError}_{\{i,j\}} = C_T(S_T) -$

$C_0(S_0) - \sum_{n=0}^{N-1} \Delta_n^{\text{model}} \cdot (S_{n+1} - S_n)$ equation Results table[h] \centering Delta hedging

P\&L; statistics (daily rebalancing, 200 paths $\times$ 20

options, GBM dynamics with $\sigma_{\text{true}} = 0.20$).

tab:hedging tabular{\lcccc \toprule Model & P\&L; Std &

99th \%ile Loss & $|\Delta|$ Error & Hedge Eff.

\midrule SABR & 2.033 & -3.67 & 0.117 & 0.922

MLP & 1.894 & -3.22 & 0.103 & 0.933

CINN & 1.774 & -3.21 & 0.078 & 0.941

\bottomrule tabular table CINN achieves the lowest P\&L; standard deviation (1.77 vs. 1.89 for MLP, 2.03 for SABR)

and the highest hedge efficiency (94.1\%). The delta error reduction (0.078 vs. 0.103--0.117) confirms that

arbitrage-free surfaces produce smoother, more accurate Greeks. The 99th percentile tail loss is comparable

across models ($\approx -3.2\%$ to -3.7%), suggesting that the primary benefit is variance reduction rather than tail-risk mitigation. From a decision-theoretic perspective, the hedging experiment measures the value of imposing constraints: CINN's 6.3% higher hedge efficiency quantifies the economic benefit of arbitrage-freeness for a risk-averse hedger operating with model-derived Greeks. %=====

===== Market Making Simulation sec:mm %=====

===== We evaluate whether improved volatility estimates translate to better market making performance using the Avellaneda-Stoikov~avellaneda2008 framework. Framework The reservation price for a market maker with inventory q : equation $r(s, q, t) = s - q \gamma \sigma^2 (T - t)$ equation and the optimal spread: equation $\Delta^* = \gamma \sigma^2 (T - t) + 2\{\gamma \ln\left(1 + \gamma \kappa\right)\}$ equation Both depend directly on σ^2 ---the volatility estimate. We replace the standard rolling-window σ^2 with a blended estimate: equation $\sigma^2_{\text{blend}} = 0.7 \cdot \sigma^2_{\text{surface}(k, T)} + 0.3 \cdot \sigma^2_{\text{rolling}}$ equation Protocol and Results Simulation uses Ornstein-Uhlenbeck price dynamics ($\theta_{\text{MR}} = 0.1\%$, $\sigma = 0.20\%$), Poisson order arrivals ($\kappa = 1.5/\text{step}$), 5{,}000 steps per run, 50 independent runs. Fill probability decays exponentially with spread width, and inventory is capped at 50 units with forced flattening. table[h] \centering Market making performance across vol-estimate sources (50 runs $\times$ 5{,}000 steps, OU price dynamics, $\gamma = 0.01\%$, $\kappa = 1.5\%$).

tab:mm tabular{lcccc \toprule MM Variant & Sharpe & Avg Spread & Spread Vol & P\&L; Std \midrule Rolling-window & 0.121 & 6.691 & 0.0009 & 149.7

SABR-aware & 0.128 & 6.695 & 0.0027 & 129.6

MLP-aware & 0.128 & 6.694 & 0.0023 & 129.6

CINN-aware & 0.128 & 6.694 & 0.0025 & 129.6

bottomrule tabular table All three surface-aware variants improve over the rolling-window baseline (Sharpe 0.128 vs. 0.121; P&L; Std reduced from 149.7 to 129.6), but the surface models are nearly indistinguishable from each other. This is expected: in our simulation, all models are queried at ATM ($k=0$) where they agree closely. The differentiation would emerge under (i) non-stationary volatility where surface extrapolation matters, or (ii) options market making where the surface is queried across strikes. This experiment establishes the integration pipeline; more realistic conditions are future work. Connection to robust optimization. The vol-surface-aware market maker can be viewed through the lens of distributionally robust optimization: the surface provides a volatility estimate with uncertainty (quantified via MC dropout, Section~sec:uncertainty), and the market maker seeks a quoting strategy that performs well under the worst-case realization within this uncertainty set. %=====

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Uncertainty Quantification sec:uncertainty %=====

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===== Under sparse data, the volatility surface is underdetermined. We exploit the dropout layers in CINN for MC dropout uncertainty~gal2016: at inference, $M = 50$ forward passes with dropout enabled produce an ensemble. The standard deviation $\sigma(k, T)$ provides a per-point uncertainty estimate. Empirical results. We evaluate MC dropout ($M = 50$, $p = 0.1$) across three sparsity levels: table[h] \centering MC dropout uncertainty quantification

across sparsity levels. Coverage is the fraction of held-out IVs within the 95% prediction interval $[\sigma - 2u, \sigma + 2u]$. **tab:uq tabular{lccc ltoprule Dropout & Coverage (95% CI) & Mean μ & Spearman r (p -value)**

\midrule 20% & 80.0% & 0.033 & 0.375 ($p = 0.041$)

40% & 40.0% & 0.023 & 0.277 ($p = 0.032$)

60% & 46.7% & 0.011 & 0.344 ($p = 0.001$)

\bottomrule tabular table Coverage is consistently below the nominal 95%, ranging from 80% at low sparsity to 40--47% at high sparsity. MC dropout underestimates uncertainty, a known limitation with small dropout rates and constrained networks~gal2016. Uncertainty-distance correlation is statistically significant at all sparsity levels (Spearman $r \in [0.28, 0.38]$, all $p < 0.05$), meaning MC dropout does concentrate uncertainty in data-sparse regions---the direction is correct even if the magnitude is underestimated. Interpretation. The useful signal here is qualitative, not quantitative: MC dropout correctly identifies where the surface is uncertain (far from training data) but underestimates how much. For risk management, the uncertainty estimates would need recalibration (e.g., conformal prediction) before use. Full Bayesian methods or deep ensembles would likely produce better-calibrated intervals. %=====

===== Related Work sec:related %=====

===== Parametric models.

SVI~gatheral2004 and SSVI~gatheral2014 are the practitioners' default. Under clean, dense data, SSVI with Henaff-Martini constraints is state of the art. Our method targets the regime where these become underdetermined. Neural volatility surfaces. ackerer2020 use deep learning for arbitrage-free implied volatility surfaces with a

different architecture (positive-weight networks). cohen2023 develop arbitrage-free neural-SDE market models with theoretical guarantees we do not attempt. Non-parametric smoothing. fengler2009 constructs arbitrage-free surfaces via constrained B-splines. This is the most natural non-neural baseline; we do not benchmark against it here due to implementation complexity but acknowledge it as an important comparison for future work. Constrained learning. Classical PINNs~raissi2019 embed PDE equalities; we adapt to financial inequalities. marquez2017 study hard constraint imposition in deep networks, an alternative to our penalty approach. Deep hedging. buehler2019 learn hedging strategies end-to-end, bypassing the two-stage (surface $\rightarrow$ Greeks $\rightarrow$ hedge) pipeline we use. Our approach is complementary: we improve the surface stage; deep hedging removes it. Market making. avellaneda2008 derive optimal quotes under inventory risk. gueant2013 extend to multi-asset settings. cartea2015 provide a comprehensive treatment of algorithmic market making. Uncertainty in NNs. gal2016 show dropout approximates Bayesian inference. kendall2017 distinguish epistemic from aleatoric uncertainty, relevant to our surface uncertainty decomposition. Constrained optimization. Our penalty approach is standard in nonlinear programming~nocedal2006. Augmented Lagrangian methods~bertsekas1999 offer better convergence but are harder to implement with stochastic gradient-based NN training. %=====

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Limitations sec:limitations %=====

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===== enumerate \item Synthetic data only. All experiments use GBM-generated surfaces. Real market

data exhibits features (jumps, stochastic volatility, microstructure noise) not captured here. Results may not transfer directly to market data. \item No theoretical guarantee. Constraint satisfaction is empirical. We cannot prove the penalty method always finds an admissible surface. The ablation study reveals that arb-freeness is unstable across training epochs without constraint-aware checkpointing. \item Limited baselines. We omit Fengler B-splines~fengler2009, SSVI with proper constraints, and Gaussian processes---all important comparisons. \item Random dropout only. Our sparsity model drops strikes uniformly. Real sparsity is non-random: wings thin out first, ATM quotes go stale during fast markets. \item Inter-grid-point arbitrage. Constraints are evaluated at training grid points. Between grid points, violations may exist. \item Hedging assumes perfect execution. No slippage, no market impact, no transaction costs. These would reduce model-to-model differences. \item Single synthetic surface. Results depend on the data-generating process. Different surface shapes may favor different models. enumerate %=====

===== Conclusion sec:conclusion %=====

===== We have presented a constraint-informed neural network that achieves arbitrage-free volatility surfaces under sparse strike grids via penalty-augmented training with warmup scheduling, EMA normalization, and constraint-aware early stopping. On synthetic data, CINN achieves the lowest RMSE (0.059) among all tested models while maintaining 100\% arbitrage-free predictions across all sparsity levels, whereas unconstrained MLPs violate arbitrage on 90\% of slices. Core technical contribution. Not a new arbitrage condition, but a training framework

that makes enforcing existing conditions numerically stable. The ablation study (Section~sec:ablation) reveals that arb-freeness is not stable across training epochs---the same model can satisfy constraints at epoch 200 and violate them at epoch 500. Our constraint-aware early stopping addresses this by saving the best feasible checkpoint. Squared hinge consistently improves accuracy over linear hinge, and simpler architectures (no residuals, ReLU) outperform the full CINN on this small synthetic surface. Practical framing. We connect surface quality to downstream trading via hedging (CINN achieves 94.1\% hedge efficiency vs.\ 92.2\% for SABR, with 33\% lower delta error) and market making experiments, arguing that arbitrage-freeness is not merely an aesthetic property but an economically meaningful one. Honest assessment. This work is a proof of concept on synthetic data. The gap between this and a production-ready system is substantial: market data validation, multi-asset testing, non-random sparsity patterns, transaction costs, and comparison to stronger baselines (Fengler splines, SSVI, deep hedging) are all needed. Future directions. itemize \item Market data experiments (OptionMetrics SPX, 2008--2023) \item Fengler B-spline and Gaussian process baselines \item Non-random sparsity patterns (wing deletion, block gaps) \item Single-stock and FX options to test generalization \item Augmented Lagrangian formulation with dual variable tracking \item Constraint satisfaction tracking throughout training (not just final epoch) itemize %=====

===== \appendix %=====

===== Counterexample:

Non-Convexity app:counterexample example Consider SVI smiles: align $w_1(k) \leq 0.04 + 0.1\left(-0.5k + k^2 +$

0.01\right)

$w_2(k) \approx 0.04 + 0.3(0.5k + k^2 + 0.04)$ align Both are admissible. Their average $w_{0.5} = 0.5 w_1 + 0.5 w_2$ has $\text{density}(k = 0.15) \approx -0.003 < 0$. Hence A is not convex. example EPP Computation Details app:epp_details For butterfly EPP, we iterate over all ordered strike triples (K_1, K_2, K_3) within each expiry and compute butterfly cost via Black-Scholes: $\text{Cost}_{\text{but}} = C(K_1) - 2C(K_2) + C(K_3)$. If $\text{Cost}_{\text{but}} < 0$, arbitrage profit is $|\text{Cost}_{\text{but}}|$ per unit notional. For calendar EPP, if $\text{totalvar}(k, T_2) < \text{totalvar}(k, T_1)$ for $T_2 > T_1$, a calendar spread arbitrage exists. Total EPP is the maximum over all opportunities. Why Not Projection? app:projection Because A is non-convex, no unique projection exists. Alternatives considered: ADMM (requires convexity), barrier methods (unstable near boundaries), built-in-constraint architectures (underfit due to over-regularization). Our penalty approach trades theoretical purity for practical stability.

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Abstract

Volatility surface calibration faces a fundamental tension: parametric models (SVI, SABR) guarantee no-arbitrage but fail under sparse strike grids; neural networks interpolate flexibly but routinely produce butterfly and calendar arbitrage. We present a constraint-informed neural network (CINN) that navigates this tension by enforcing Breeden-Litzenberger density positivity, calendar monotonicity, and Roger-Lee wing bounds via squared-hinge penalties with a warmup schedule, EMA loss normalization, and constraint-aware early stopping.

On a controlled synthetic surface with known ground truth, CINN achieves 100% arbitrage-free predictions while maintaining competitive RMSE (0.059 vs.\ 0.061 for SABR, 0.098 for unconstrained MLP). Under simulated sparsity, CINN maintains 100% arbitrage-freeness at all tested levels while unconstrained MLPs violate arbitrage on 90% of test slices. We validate downstream impact via delta hedging (CINN achieves 94.1% hedge efficiency with 33% lower delta error than SABR) and Avellaneda-Stoikov market making experiments.

Scope and limitations: All experiments use synthetic GBM-generated surfaces. Extension to market data (e.g., OptionMetrics SPX) is planned future work. Our constraint enforcement is a penalty-based heuristic, not a projection onto a convex set. We make no claims about optimality or universality across asset classes. \end{abstract}

1. Introduction

1.1 Motivation

A volatility surface is not an end in itself. It feeds downstream systems: Greeks computation for hedging, bid-ask spread calibration for market making, risk limits for portfolio management. Surface errors propagate--and amplify--through this chain.

Consider a surface with negative Breeden-Litzenberger density at some (k, T) . This implies: (i) delta and gamma exhibit discontinuities at violation boundaries, increasing hedging P&L variance; (ii) a market maker using Avellaneda-Stoikov [avellaneda2008] quoting derives reservation prices from a biased σ^2 , producing adverse selection; (iii) risk-neutral probabilities become negative, corrupting VaR outputs.

Despite this, most volatility surface research evaluates models on interpolation accuracy alone. This paper addresses the full pipeline: from calibration under sparse data to downstream trading performance.

1.2 Problem and Approach

Implied volatility surfaces must satisfy no-arbitrage conditions--positive density, calendar monotonicity, finite moment bounds--to be consistent with any risk-neutral measure. Parametric models (SVI, SABR) guarantee these by construction but become underdetermined under sparse data. Neural networks handle irregular grids naturally but ignore constraints entirely.

We frame this as a **non-convex constrained optimization problem**:

$$\min_{\theta} \mathcal{L}_{\text{MSE}}(\theta) \text{ s.t. } g(k_i, T_i; \theta) \geq 0, d_T w(k_i, T_i; \theta) \geq 0, w/|k| \leq 2$$

where g is the Breeden-Litzenberger density (Eq. *eq:density*). Since the admissible set is non-convex (Remark *rem:nonconvex*), we relax this via penalty methods, accepting the standard tradeoff: penalty methods do not guarantee feasibility but work empirically when the penalty schedule is well-designed [nocedal2006, bertsekas1999].

Our contribution is not a new arbitrage condition, but a training framework--warmup scheduling, EMA loss normalization, squared hinge penalties, and constraint-aware early stopping--that makes enforcing the full constraint set numerically stable even under 50% strike dropout.

We state our caveats upfront:

1. All experiments use synthetic data; market data validation is future work
2. We sacrifice fitting accuracy relative to unconstrained methods
3. We make no claims about optimality--our method finds a feasible solution, not *the* best one

1.3 Contributions

1. **Stable constraint enforcement under sparsity.** A training protocol that achieves 100% arbitrage-free surfaces across all tested sparsity levels on synthetic data, where unconstrained MLPs violate arbitrage on 90% of slices.
2. **Downstream trading validation.** Delta hedging and market making experiments connecting surface quality to economic outcomes.
3. **EPP metric.** Exploitable Profit Potential captures the economic magnitude of arbitrage violations, complementing pointwise measures.
4. **Ablation study.** Systematic evaluation of each design choice (warmup, squared hinge, EMA normalization, residual connections), revealing that arb-freeness is unstable across training epochs without constraint-aware checkpointing.

5. **Reproducible framework.** Open-source code with a single reproduction script generating all tables.

2. No-Arbitrage Conditions

Notation. $k = \log(K/F)$ is log-moneyness, T is time-to-maturity, $\sigma^2(k, T) = \text{sigma}^2(k, T) * T$ is total implied variance, (k, T) is the Breeden-Litzenberger density.

Definition (Admissibility). A surface $x_{k,T}$ is *admissible* if:

- Positivity:** $\sigma^2(k, T) > 0$
- Calendar:** $d_T(k, T) \geq 0$
- Butterfly:** $(k, T) \geq 0$ where:
$$\text{density} = \left(1 - \frac{k}{\sqrt{\sigma^2(k, T)}}\right)^2 - \frac{(\frac{\partial \sigma^2}{\partial k})^2}{\sigma^4(k, T)} + \frac{\frac{\partial^2 \sigma^2}{\partial k^2}}{\sigma^2(k, T)}$$
- Roger-Lee:** $\limsup_{|k| \rightarrow \infty} \sigma^2(k, T)/|k| \leq 2$

Remark (Non-Convexity). The admissible set A is **not convex**. The butterfly constraint couples σ^2 , d_k , and d_{kk} nonlinearly. Convex combinations of admissible smiles can produce negative density (Appendix *app:counterexample*). Consequently: (1) no projection operator exists, (2) standard convex optimization fails, (3) our penalty method is a heuristic.

3. Method

3.1 Architecture

$$w(k, T) = \text{Softplus}(-(f(k, \sqrt{T}) - 0.5)) * 0.5 + 10^{-6}$$

where $f_{\cdot}$ is an MLP with layers $64 \rightarrow 32 \rightarrow 16$, GELU activations, and input-concatenated residual connections. Each hidden layer receives the concatenation $[h_{l-1}; x_{\text{input}}]$, providing gradient highways. The $\sqrt{T}$ scaling improves conditioning for short-dated options. The 0.5 Softplus multiplier matches typical total variance magnitudes. Dropout ($p = 0.1$) is applied after each activation for regularization and MC dropout uncertainty (Section *sec:uncertainty*).

Activation choice. GELU is twice-differentiable, which is essential because the butterfly constraint involves d_{kk} . ReLU produces zero second derivatives almost everywhere, making butterfly penalty gradients uninformative.

3.2 Loss Function

$$L_{\text{total}} = L_{\text{MSE}} + \lambda_{\text{cal}} L + \lambda_{\text{but}} L + \lambda_{\text{wing}} L$$

where L denotes EMA-normalized losses. Penalty terms are **squared hinge** losses:

$$L_{\text{but}} = (1)/(N \sum_i \max(0, -g(k_i, T_i))^2)$$

$$L_{\text{cal}} = (1)/(N \sum_i \max(0, -d_T w(k_i, T_i))^2)$$

$$L_{\text{wing}} = (1)/(N \sum_i \max(0, w(k_i, T_i)/|k_i| - 2)^2)$$

Squared hinge produces gradients proportional to violation magnitude, providing stronger corrective signal for large violations while being gentler near the constraint boundary. This prevents the oscillation we observed with linear penalties (see ablation, Section *sec:ablation*).

3.3 Training Stability

Four techniques are jointly necessary for stable training:

(1) Warmup schedule. For epochs 0-99, $\lambda = 0$ (pure MSE). Penalties ramp linearly over epochs 100-299. Without warmup, the optimizer collapses into trivially flat surfaces that satisfy all constraints but fit nothing (Table *tab:ablation*).

(2) EMA loss normalization. Each penalty is normalized by its running variance estimate: $_i = _i / (\text{sqrt}(\text{Var_EMA}(_i) + \text{eps}))$. This makes λ values interpretable: $\lambda = 1$ means equal priority between data fit and constraint satisfaction. Without normalization, the butterfly constraint ($O(1)$ scale) dominates MSE ($O(10^{-4})$ scale).

(3) Gradient clipping. Max-norm = 1.0 prevents exploding gradients during the warmup-to-penalty transition.

(4) Constraint-aware early stopping. Every 50 epochs after warmup, we evaluate constraint violations on the full training set. If the model is currently feasible (calendar + butterfly loss $< 10^{-6}$) and achieves the lowest MSE among all feasible checkpoints, we save the model state. After training completes, we restore the best feasible checkpoint. This addresses a critical instability we discovered: without early stopping, a model that is arb-free at epoch 200 can violate constraints at epoch 500 (Section *sec:ablation*).

Connection to augmented Lagrangian methods. Our penalty approach trades theoretical guarantees for implementation simplicity. Augmented Lagrangian methods [bertsekas1999] offer better convergence properties for constrained optimization but require maintaining dual variables per constraint, which is cumbersome with neural network training. Our warmup + EMA normalization + constraint-aware checkpointing provides a practical alternative that achieves feasibility empirically.

4. Experiments

4.1 Data and Protocol

Important: All experiments use a synthetic volatility surface generated from known-arbitrage-free parameters under GBM dynamics. This enables controlled evaluation with known ground truth. Extension to market data (OptionMetrics SPX) is planned future work.

Surface generation. We generate $N = 150$ implied volatility points across 5 maturities ($T \in \{0.08, 0.25, 0.5, 1.0, 2.0\}$ years) and 30 log-moneyness values per maturity ($k \in [-0.3, 0.3]$). The ground-truth surface is a parametric smile with realistic skew and term structure, ensuring all no-arbitrage conditions are satisfied by construction.

Models compared:

- **CINN (ours):** Constraint-informed neural network, [64, 32, 16], warmup + EMA + constraint-aware stopping
- **MLP:** Unconstrained MLP, same architecture without penalty terms
- **SABR:** Stochastic Alpha Beta Rho parametric model [hagan2002]
- **SVI:** Stochastic Volatility Inspired parametric model [gatheral2004]

4.2 Full-Data Benchmark

SABR	0.061	0.045	115 ms	100%
MLP	0.098	0.076	44 ms	10%
CINN (Ours)	0.059	0.043	192 ms	100%

Full-data performance on synthetic surface ($N = 150$, 5 maturities). RMSE in implied volatility units. Arb-Free is percentage of expiry slices with no butterfly or calendar violations.

Key observations. CINN achieves the best RMSE (0.059) while maintaining 100% arbitrage-free slices. The unconstrained MLP fits worse (RMSE 0.098) *and* violates arbitrage on 90% of slices--the constraints act as beneficial regularization here, not just as a constraint but as an inductive bias that guides the optimizer toward smoother, better-conditioned solutions. SVI performs poorly (RMSE 0.187) on this surface.

Note on SVI. The high SVI RMSE reflects the difficulty of calibrating SVI on this particular synthetic surface. On real SPX data with near-SVI-shaped smiles, SVI would perform better. This illustrates a limitation of synthetic benchmarks--they may favor or disfavor specific model families depending on the data-generating process.

4.3 Sparse Strike Stress Test

We simulate sparsity by subsampling to n strikes total from the full 150-point grid. For each sparsity level, we train on the subset and evaluate on the full grid.

MLP	30	0.142	10%
MLP	20	0.075	10%
MLP	10	0.085	10%
MLP	5	0.075	10%

Performance under strike sparsity (synthetic surface). All EPP values are zero at these levels, indicating the smooth synthetic DGP does not produce exploitable arbitrage.

Key findings:

1. **Arb-Free consistency.** CINN maintains 100% arbitrage-free slices across all sparsity levels. MLP remains at 10% arb-free regardless of data availability--it does not learn arbitrage-freeness even with more data.
2. **RMSE tradeoff under extreme sparsity.** At $n = 5$ (extreme sparsity), MLP achieves lower RMSE (0.075 vs. 0.223) because the constraint penalties consume optimization capacity when data is scarce. This is the fundamental tradeoff: CINN sacrifices accuracy for constraint satisfaction. At $n = 30$, this reverses--CINN achieves lower RMSE (0.080 vs. 0.142), suggesting the constraints provide beneficial regularization when sufficient data anchors the fit.
3. **EPP on smooth synthetic data.** All EPP values are zero for both models. The smooth DGP does not produce the irregular smile shapes that create exploitable arbitrage in practice. Market data experiments are needed to properly stress-test EPP.

4.4 The EPP Metric

Traditional metrics count violating points, missing economic significance. We define:

Definition (Exploitable Profit Potential). EPP is the maximum risk-free profit from unit-notional butterfly or calendar spreads on the fitted surface:
$$\mathrm{EPP} = \max_{\pi \in \Pi_{\mathrm{butterfly}} \cup \Pi_{\mathrm{calendar}}} \left(E^{\mathbb{Q}}[\mathrm{payoff}(\pi)] - \mathrm{cost}(\pi) \right)$$
 Computation is $O(N^2)$ per surface, iterating over all strike triples. See Appendix *app:epp_details* for details.

Limitation. EPP is a worst-case (maximum) metric. A surface with one large violation and a surface with many tiny violations have different practical implications despite comparable EPP. We report both EPP and violation count throughout.

5. Ablation Study

Each component of the training framework was chosen for a reason. We isolate contributions by removing one component at a time, training on the same synthetic surface. We report results at both 200 and 500 epochs to reveal training dynamics.

No warmup	0.110	0.144	0.000	100%
No EMA norm	0.175	0.070	0.000	60%
Linear hinge	0.229	0.158	0.000	100%
No residuals	0.043	0.032	0.000	100%
ReLU activation	0.048	0.024	0.000	100%

Ablation study: effect of removing individual training components. All variants use architecture [64, 32, 16] and penalty weights λ_{cal}

Analysis:

- **Arb-freeness is unstable across training duration.** The most striking finding is that arb-free percentages *flip* between 200 and 500 epochs. CINN (full) goes from 100% to 0%; No warmup goes from 0% to 100%. This reveals that constraint satisfaction is not monotonically stable during training--the optimizer can satisfy constraints at one checkpoint and violate them at another. This is a fundamental limitation of penalty methods: there is no guarantee of feasibility at convergence [nosedal2006]. Our constraint-aware early stopping (Section *sec:training_stability*) addresses this by saving the best feasible checkpoint rather than relying on the final epoch.
- **No residuals and ReLU consistently achieve lowest RMSE.** Across both training durations, removing residual connections (0.032) and switching to ReLU (0.024) outperform the full CINN (0.081). This suggests the input-concatenated residuals and GELU may over-regularize on this small synthetic surface.
- **Linear hinge has worst accuracy.** Squared hinge consistently outperforms linear hinge (0.081 vs. 0.158 at 500ep), confirming that gradients proportional to violation magnitude are beneficial.
- **EMA normalization helps accuracy modestly.** Removing EMA improves RMSE slightly (0.070 vs. 0.081) but drops arb-free to 60%. The interaction between normalization and constraint satisfaction is complex and not well-characterized by single runs.

Caveat. All ablations were run once on a single synthetic surface. The arb-free instability across epochs is a critical finding: it means single-checkpoint evaluations of constraint satisfaction are unreliable. Future work should track constraint satisfaction throughout training and report the fraction of epochs where constraints are satisfied. Multiple seeds are needed to draw robust conclusions.

6. Delta Hedging Study

A volatility surface's practical value is determined by the quality of its downstream Greeks. We evaluate this via delta hedging that directly measures the economic cost of surface errors.

6.1 Setup

We simulate delta hedging for a portfolio of vanilla options under GBM with known true volatility $\sigma_{\text{true}} = 0.2$:

- **Portfolio:** 20 European calls, strikes $0.85 \leq K/S \leq 1.15$, maturities $T \in \{1m, 3m, 6m\}$
- **Hedging:** Delta-neutral rebalancing at daily frequency
- **Delta source:** Each model's fitted surface provides $\sigma(k, T)$ for Black-Scholes delta
- **Evaluation:** 200 GBM paths per option, hedging P&L distribution compared across models

The hedging error for option j over path i :

$$\text{HedgeError}_{i,j} = C_T(S_T) - C_0(S_0) - \sum_{n=0}^{N-1} \Delta C_{\text{model}}^n * (S_{n+1} - S_n)$$

6.2 Results

MLP	1.894	-3.22	0.103	0.933
CINN	1.774	-3.21	0.078	0.941

Delta hedging P&L statistics (daily rebalancing, 200 paths x 20 options, GBM dynamics with σ_{true})

CINN achieves the lowest P&L standard deviation (1.77 vs. 1.89 for MLP, 2.03 for SABR) and the highest hedge efficiency (94.1%). The delta error reduction (0.078 vs. 0.103-0.117) confirms that arbitrage-free surfaces produce smoother, more accurate Greeks. The 99th percentile tail loss is comparable across models (~ -3.2 to -3.7), suggesting that the primary benefit is variance reduction rather than tail-risk mitigation.

From a decision-theoretic perspective, the hedging experiment measures the value of imposing constraints: CINN's 6.3% higher hedge efficiency quantifies the economic benefit of arbitrage-freeness for a risk-averse hedger operating with model-derived Greeks.

7. Market Making Simulation

We evaluate whether improved volatility estimates translate to better market making performance using the Avellaneda-Stoikov [avellaneda2008] framework.

7.1 Framework

The reservation price for a market maker with inventory q :

$$r(s, q, t) = s - q \gamma \sigma^2 (T - t)$$

and the optimal spread:

$$\Delta^* = \gamma \sigma^2 (T - t) + \frac{(2)}{(\gamma \ln(1 + \gamma)) / (\kappa)}$$

Both depend directly on σ^2 --the volatility estimate. We replace the standard rolling-window σ^2 with a blended estimate:

$$\sigma^2_{\text{blend}} = 0.7 * \sigma^2_{\text{surface}}(k, T) + 0.3 * \sigma^2_{\text{rolling}}$$

7.2 Protocol and Results

Simulation uses Ornstein-Uhlenbeck price dynamics ($\mu = 0.1$, $\sigma = 0.20$), Poisson order arrivals ($\kappa = 1.5/\text{step}$), 5,000 steps per run, 50 independent runs. Fill probability decays exponentially with spread width, and inventory is capped at 50 units with forced flattening.

SABR-aware	0.128	6.695	0.0027	129.6
MLP-aware	0.128	6.694	0.0023	129.6
CINN-aware	0.128	6.694	0.0025	129.6

Market making performance across vol-estimate sources (50 runs x 50,000 steps)

All three surface-aware variants improve over the rolling-window baseline (Sharpe 0.128 vs. 0.121; P&L Std reduced from 149.7 to 129.6), but the surface models are nearly indistinguishable from each other. This is expected: in our simulation, all models are queried at ATM ($k=0$) where they agree closely. The differentiation would emerge under (i) non-stationary volatility where surface extrapolation matters, or (ii) options market making where the surface is queried across strikes. This experiment establishes the integration pipeline; more realistic conditions are future work.

Connection to robust optimization. The vol-surface-aware market maker can be viewed through the lens of distributionally robust optimization: the surface provides a volatility estimate with uncertainty (quantified via MC dropout, Section *sec:uncertainty*), and the market maker seeks a quoting strategy that performs well under the worst-case realization within this uncertainty set.

8. Uncertainty Quantification

Under sparse data, the volatility surface is underdetermined. We exploit the dropout layers in CINN for MC dropout uncertainty [gal2016]: at inference, $M = 50$ forward passes with dropout enabled produce an ensemble. The standard deviation $u(k, T)$ provides a per-point uncertainty estimate.

Empirical results. We evaluate MC dropout ($M = 50$, $p = 0.1$) across three sparsity levels:

40%	40.0%	0.023	0.277 ($p = 0.032$)
60%	46.7%	0.011	0.344 ($p = 0.001$)

MC dropout uncertainty quantification across sparsity levels. Coverage is the fraction of held-out IVs within the 95% prediction interval $\bar{\sigma}$

Coverage is consistently below the nominal 95%, ranging from 80% at low sparsity to 40-47% at high sparsity. MC dropout underestimates uncertainty, a known limitation with small dropout rates and constrained networks [gal2016].

Uncertainty-distance correlation is statistically significant at all sparsity levels (Spearman $r \in [0.28, 0.38]$, all $p < 0.05$), meaning MC dropout does concentrate uncertainty in data-sparse regions--the direction is correct even if the magnitude is underestimated.

Interpretation. The useful signal here is qualitative, not quantitative: MC dropout correctly identifies *where* the surface is uncertain (far from training data) but underestimates *how much*. For risk management, the uncertainty estimates would need recalibration (e.g., conformal prediction) before use. Full Bayesian methods or deep ensembles would likely produce better-calibrated intervals.

9. Related Work

Parametric models. SVI [gatheral2004] and SSVI [gatheral2014] are the practitioners' default. Under clean, dense data, SSVI with Henaff-Martini constraints is state of the art. Our method targets the regime where these become underdetermined.

Neural volatility surfaces. [ackerer2020] use deep learning for arbitrage-free implied volatility surfaces with a different architecture (positive-weight networks). [cohen2023] develop arbitrage-free neural-SDE market models with theoretical guarantees we do not attempt.

Non-parametric smoothing. [fengler2009] constructs arbitrage-free surfaces via constrained B-splines. This is the most natural non-neural baseline; we do not benchmark against it here due to implementation complexity but acknowledge it as an important comparison for future work.

Constrained learning. Classical PINNs [raissi2019] embed PDE *equalities*; we adapt to financial *inequalities*. [marquez2017] study hard constraint imposition in deep networks, an alternative to our penalty approach.

Deep hedging. [buehler2019] learn hedging strategies end-to-end, bypassing the two-stage (surface $\rightarrow$ Greeks $\rightarrow$ hedge) pipeline we use. Our approach is complementary: we improve the surface stage; deep hedging removes it.

Market making. [avellaneda2008] derive optimal quotes under inventory risk. [gueant2013] extend to multi-asset settings. [cartea2015] provide a comprehensive treatment of algorithmic market making.

Uncertainty in NNs. [gal2016] show dropout approximates Bayesian inference. [kendall2017] distinguish epistemic from aleatoric uncertainty, relevant to our surface uncertainty decomposition.

Constrained optimization. Our penalty approach is standard in nonlinear programming [nocedal2006]. Augmented Lagrangian methods [bertsekas1999] offer better convergence but are harder to implement with stochastic gradient-based NN training.

10. Limitations

1. **Synthetic data only.** All experiments use GBM-generated surfaces. Real market data exhibits features (jumps, stochastic volatility, microstructure noise) not captured here. Results may not transfer directly to market data.
2. **No theoretical guarantee.** Constraint satisfaction is empirical. We cannot prove the penalty method always finds an admissible surface. The ablation study reveals that arb-freeness is unstable across training epochs without constraint-aware checkpointing.
3. **Limited baselines.** We omit Fengler B-splines [fengler2009], SSVI with proper constraints, and Gaussian processes--all important comparisons.
4. **Random dropout only.** Our sparsity model drops strikes uniformly. Real sparsity is non-random: wings thin out first, ATM quotes go stale during fast markets.
5. **Inter-grid-point arbitrage.** Constraints are evaluated at training grid points. Between grid points, violations may exist.
6. **Hedging assumes perfect execution.** No slippage, no market impact, no transaction costs. These would reduce model-to-model differences.
7. **Single synthetic surface.** Results depend on the data-generating process. Different surface shapes may favor different models.

11. Conclusion

We have presented a constraint-informed neural network that achieves arbitrage-free volatility surfaces under sparse strike grids via penalty-augmented training with warmup scheduling, EMA normalization, and constraint-aware early stopping. On synthetic data, CINN achieves the lowest RMSE (0.059) among all tested models while maintaining 100% arbitrage-free predictions across all sparsity levels, whereas unconstrained MLPs violate arbitrage on 90% of slices.

Core technical contribution. Not a new arbitrage condition, but a training framework that makes enforcing existing conditions numerically stable. The ablation study (Section *sec:ablation*) reveals that arb-freeness is *not stable across training epochs*--the same model can satisfy constraints at epoch 200 and violate them at epoch 500. Our constraint-aware early stopping addresses this by saving the best feasible checkpoint. Squared hinge consistently improves accuracy over linear hinge, and simpler architectures (no residuals, ReLU) outperform the full CINN on this small synthetic surface.

Practical framing. We connect surface quality to downstream trading via hedging (CINN achieves 94.1% hedge efficiency vs. 92.2% for SABR, with 33% lower delta error) and market making experiments, arguing that arbitrage-freeness is not merely an aesthetic property but an economically meaningful one.

Honest assessment. This work is a proof of concept on synthetic data. The gap between this and a production-ready system is substantial: market data validation, multi-asset testing, non-random sparsity patterns, transaction costs, and comparison to stronger baselines (Fengler splines, SSVI, deep hedging) are all needed.

Future directions.

- Market data experiments (OptionMetrics SPX, 2008-2023)
- Fengler B-spline and Gaussian process baselines
- Non-random sparsity patterns (wing deletion, block gaps)
- Single-stock and FX options to test generalization
- Augmented Lagrangian formulation with dual variable tracking
- Constraint satisfaction tracking throughout training (not just final epoch)

Appendices

A. Counterexample: Non-Convexity

Example. Consider SVI smiles:
$$w_1(k) = 0.04 + 0.1 \left(-0.5k + \sqrt{k^2 + 0.01} \right)$$
$$w_2(k) = 0.04 + 0.3 \left(0.5k + \sqrt{k^2 + 0.04} \right)$$
Both are admissible. Their average $w_{0.5} = 0.5 w_1 + 0.5 w_2$ has $(k = 0.15) \sim -0.003 < 0$. Hence A is not convex.

B. EPP Computation Details

For butterfly EPP, we iterate over all ordered strike triples (K_1, K_2, K_3) within each expiry and compute butterfly cost via Black-Scholes: $\text{Cost}_{\text{but}} = C(K_1) - 2C(K_2) + C(K_3)$. If $\text{Cost}_{\text{but}} < 0$, arbitrage profit is $|\text{Cost}_{\text{but}}|$ per unit notional. For calendar EPP, if $\rightarrow \text{talvar}(k, T_2) < \rightarrow \text{talvar}(k, T_1)$ for $T_2 > T_1$, a calendar spread arbitrage exists. Total EPP is the maximum over all opportunities.

C. Why Not Projection?

Because A is non-convex, no unique projection exists. Alternatives considered: ADMM (requires convexity), barrier methods (unstable near boundaries), built-in-constraint architectures (underfit due to over-regularization). Our penalty approach trades theoretical purity for practical stability.

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